
A Multi-Physics Hybrid Dynamical System for Accurately Predicting Smartphone Battery SOC Evolution and Time-to-Empty

Summary

With the rapidly increasing capabilities of smartphones, the unpredictability of battery life has become a critical pain point affecting user experience. Battery depletion is not a simple linear process but a complex nonlinear system deeply coupled with user behavior (e.g., screen brightness, CPU load), network conditions, and thermodynamic factors (e.g., temperature, aging). Addressing the problem requirements, this paper establishes a physics-based **Multi-Physics Hybrid Dynamical System** to accurately predict the State of Charge (SOC) evolution and time-to-empty under various operating conditions.

For **Problem 1**, we constructed the core architecture of the system, which consists of two coupled sub-models: the **Internal Battery Dynamics Model (Model 1)** and the **External Dynamic Load Model (Model 2)**. First, to characterize transient voltage response, we discarded simple Coulomb counting and employed a first-order **Thevenin Equivalent Circuit** combined with the **Arrhenius equation** to establish a nonlinear mapping between temperature, effective capacity, and internal resistance. To capture time-varying characteristics, the **FFRLS** algorithm was introduced for online parameter identification. Second, to accurately reconstruct the load current $I(t)$, we decomposed total power consumption into five modules: Screen (Gamma nonlinear model), Processor (DVFS dynamic model), GPS, Network (Signal strength state machine), and Background processes (Stochastic Poisson process). These two sub-models are closed-loop coupled via a **Lumped Parameter Thermal Balance Equation**: current generates Joule heating which alters temperature, and temperature conversely affects internal resistance and voltage, thereby forming a continuous dynamical system evolving over time.

For **Problem 2**, we simulated time-to-empty under distinct scenarios to identify specific failure drivers. The model reveals that **thermal-induced shutdown** dominates in high-load gaming (Scenario A), while **impedance-driven voltage collapse (UVLO)** triggers premature shutdown in cold environments (Scenario B), successfully explaining the observed "sudden death" phenomenon. **Monte Carlo simulations** were employed to quantify uncertainty, yielding a 95% confidence interval of **[0.02, 2.61] hours** for time-to-empty under extreme cold conditions. This quantitative result demonstrates that battery endurance in low temperatures is dominated by stochastic high-energy pulses rather than simple remaining capacity.

For **Problem 3**, **Sensitivity Analysis** was conducted. By constructing the **Normalized Local Sensitivity Index (NLSI)** and dual-parameter heatmaps, we identified environmental temperature and load intensity as the most critical factors affecting system stability. The heatmaps clearly delineated two high-risk domains: the "Impedance Limited Zone" (sudden death due to high load in cold) and the "Thermal Shutdown Zone" (overheating due to high load in heat), revealing the physical boundaries of safe battery operation.

Finally, based on the model conclusions, we proposed three practical recommendations for users: reducing brightness by leveraging the nonlinear power characteristics of screens, avoiding high-power operations in extreme cold to prevent voltage collapse, and enabling flight mode in weak signal areas. Additionally, we suggest that operating systems introduce temperature-aware dynamic power limiting strategies to extend the battery's full lifecycle.

Keywords: Lithium-Ion Battery; Thevenin Equivalent Circuit; Multi-Physics Coupling; Thermodynamic Feedback; Stochastic Poisson Process; Monte Carlo Simulation.

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1 Introduction

1.1 Problem Background

The rapid advancement of science and technology, smartphones have become widely adopted and deeply integrated into human society as essential tools for production and daily life, playing an irreplaceable role across multiple dimensions including work, study, and entertainment. However, as users' dependence on smartphones continues to deepen, demands for their battery endurance are also rising. Modern smartphone battery performance exhibits significant uncertainty: during daily use, battery consumption depends not only on explicit factors like screen brightness and processor load but also on complex influences such as network activity, background application processes, and ambient temperature.^[1] Traditional power estimation methods struggle to meet the demand for precise prediction.

Therefore, developing a mechanism-based model grounded in continuous-time differential equations is imperative. Given that lithium-ion batteries are a core component of smartphones^[2], our model should be theoretically grounded in the electrochemical kinetics of lithium-ion batteries. It should establish differential equations describing the evolution of the State of Charge (SOC) over time, systematically integrating the dynamic characteristics of major power-consuming modules such as screen display, processor operations, and network communication, while accounting for time-varying parameters like temperature effects and battery aging. Establishing such a physical model not only enhances the accuracy of power prediction but also provides a theoretical basis for optimizing operating system energy consumption. Ultimately, this will advance smartphone energy management toward more refined and intelligent approaches.

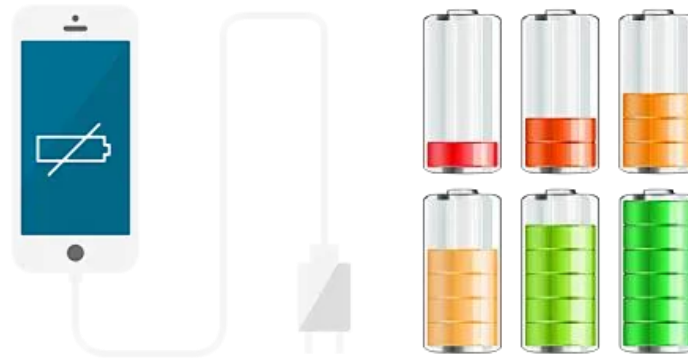


Figure 1-1 Mobile Phone Charging Power Chart

1.2 Restatement of the Problem

Against this backdrop, we plan to predict smartphone battery depletion times through data-driven mathematical modeling, addressing the following key challenges:

- Task 1: Develop an observational model for internal battery states and physical parameters: Establish a dynamic model capable of real-time acquisition of core internal physical parameters (e.g., polarization voltage, internal resistance) and state of charge (SOC) to characterize the battery's internal state.

- Task 2.:Develop a dynamic capacity degradation correction model: Establish a model that continuously tracks the dynamic decline of battery capacity over time, temperature, and charge/discharge cycles, enabling real-time correction of effective capacity.
- Task 3.:Create a user-behavior-driven power consumption decomposition model: Build a model capable of decomposing the power consumption contributions from different user operations (screen, processor, network, etc.), quantifying the power consumption characteristics of each behavior.
- Task 4: Develop a battery runtime prediction model across multiple scenarios: Predict battery runtime under varying initial charge levels across different usage scenarios and analyze key influencing factors.

1.3 Our Work

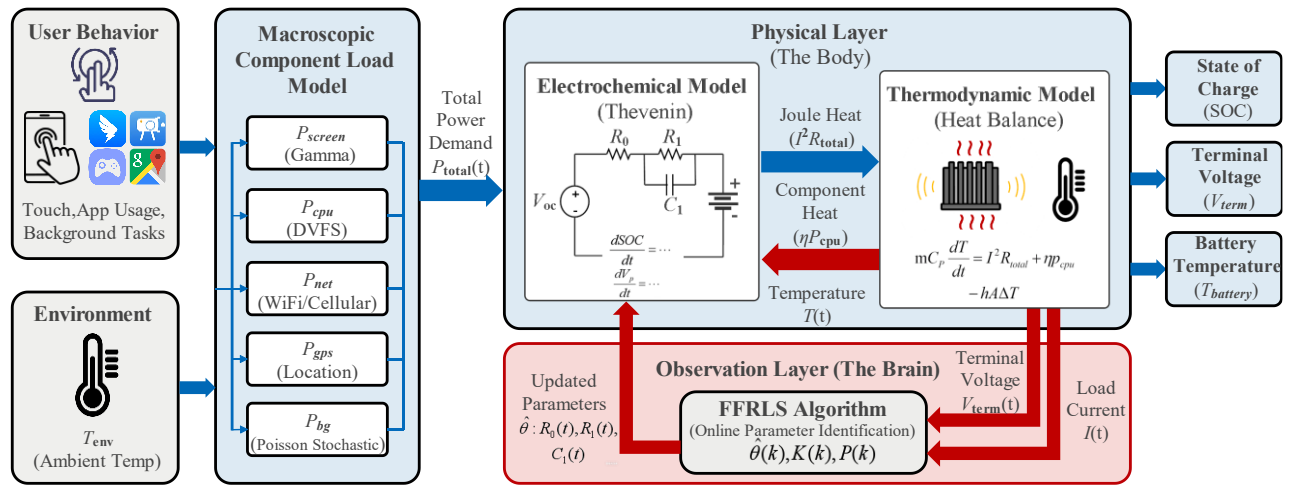


Figure 1-2 The Proposed Dual-Layer Adaptive Architectun System Diagram

2 Assumptions and Justifications

To simplify the problem, we make the following assumptions, each of which is well-founded.

- **Assumption 1: The battery is treated as a lumped-parameter thermal system with uniform internal temperature distribution.**

Justification: While microscopic thermal gradients exist within the battery, the internal temperature difference is negligible compared to the gradient between the battery and the environment, given the smartphone's compact physical scale and high thermal conductivity packaging. This allows us to model thermal equilibrium using Ordinary Differential Equations (ODEs) rather than Partial Differential Equations (PDEs), significantly reducing computational complexity while maintaining sufficient engineering accuracy.

- **Assumption 2: The battery's transient voltage response is sufficiently characterized by a first-order Thevenin equivalent circuit.**

Justification: Compared to highly complex electrochemical models (e.g., the P2D model), a first-order RC circuit is sufficient to capture millisecond-to-second scale voltage recovery and polarization effects, representing the optimal balance between computational efficiency and model

accuracy. While second-order or higher-order RC networks could theoretically improve precision, they would significantly increase the computational burden for parameter identification (risking overfitting) with negligible practical contribution to predicting the macroscopic metric of "time-to-empty."

- **Assumption 3: The internal resistance and reaction rates follow the Arrhenius Law strictly within the operating range.**

Justification: We assume that temperature is the dominant environmental factor affecting the battery's electrochemical characteristics. The variation of internal resistance with temperature strictly follows the Arrhenius relationship (i.e., resistance increases exponentially as temperature decreases). This physical assumption is critical for reproducing "Voltage Collapse" and "False Death" phenomena in low-temperature environments and is central to explaining the failure mechanism in Scenario B.

- **Assumption 4: Background application activities follow a homogeneous Poisson process.**

Justification: The occurrence of background tasks (e.g., push notifications, system synchronization) is generally memoryless, random, and independent. On a macroscopic time scale, the arrival intervals of these discrete events follow an exponential distribution, consistent with the statistical properties of a Poisson process. This provides a solid statistical foundation for using Monte Carlo simulations to quantify the uncertainty of "time-to-empty."

- **Assumption 5: The State of Health (SOH) remains quasi-static during a single discharge cycle.**

Justification: Battery aging is a prolonged process spanning months or even years. Within the time scale of a single discharge cycle (typically a few hours to a day), we consider the battery's maximum available capacity C_{max} and aging factors to be relatively constant. This decouples the "fast time scale (discharge dynamics)" from the "slow time scale (capacity fade)," simplifying the simulation model for a single discharge event.

3 Notations

The key symbols used in this paper are listed in Table 3-1.

Table 3-1 Notations used in this paper

Symbols	Definition	Units
$SOC(t)$	Battery State of Charge	/
$V_p(t)$	Polarization voltage	V
$T(t)$	Equivalent Mean Temperature	°C
$P_{total}(t)$	Instantaneous total power	W
T_{env}	Ambient temperature	°C
$I(t)$	Battery discharge current	A

C_{eff}	Effective capacity	Ah
$R_0(t)$	Ohmic Resistance	R
$R_1(t)$	Polarization Resistance	R
$C_1(t)$	Polarization Capacitance	C

*Some variables not listed in this paper will be discussed in detail below.

4 Model Preparation

4.1 State Variables and Input

The system employs lumped parameter modeling, defining the state vector as:

$$X(t) = [SOC(t), V_p(t), T(t)]^T \quad (4.1)$$

- $SOC(t)$: Battery State of Charge (SOC), current available capacity/nominal battery capacity, with a value range of : $0 \leq SOC(t) \leq 1$;
- $V_p(t)$: Polarization voltage across the RC polarization branch in the equivalent circuit model;
- $T(t)$: The equivalent average temperature of the battery and the interior of the phone.

Driven by the aforementioned state variables, the system is influenced by the following external inputs:

- $P_{total}(t)$: The instantaneous total power demand triggered by user behavior comprehensively reflects load characteristics such as screen display, processor computation, and background tasks.
- T_{env} : Ambient temperature, treated as an external constant, characterizes the influence of the surrounding thermal environment on the heat dissipation process of equipment.

4.2 Data Source

This article relies on the following table of data sources.

Table 4-1 Date Source Websites

Database names	Database website
AndroWatts ^[3]	https://zenodo.org/records/14314943
NASA Dataset ^[4]	https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/

5 Model I :Core Dynamics

5.1 Battery Dynamics

To characterize voltage sags and polarization effects, this paper employs a more accurate first-order Thevenin equivalent circuit model.

State equation set:

$$\begin{cases} \frac{dSOC(t)}{dt} = -\frac{I(t)}{3600 \cdot C_{\text{eff}}(T, \text{aging})} \\ \frac{dV_p(t)}{dt} = -\frac{V_p(t)}{R_1(T)C_1(T)} + \frac{I(t)}{C_1(T)} \end{cases} \quad (5.1)$$

- $I(t)$: Battery discharge current (A);
- C_{eff} : Effective capacity (Ah), subject to variation with temperature and aging;
- R_1, C_1 : Polarization branch parameters, reflecting electrochemical kinetic characteristics.

5.2 Output Equation (Terminal Voltage):

$$V_{\text{term}}(t) = OCV(SOC) - V_p(t) - I(t)R_0(T) \quad (5.2)$$

Among these, $OCV(SOC)$ is the open-circuit voltage function, regarded as a quasi-static function of SOC . Its time scale of variation is significantly longer than the RC polarization dynamics, and thus it is not introduced as a state variable.

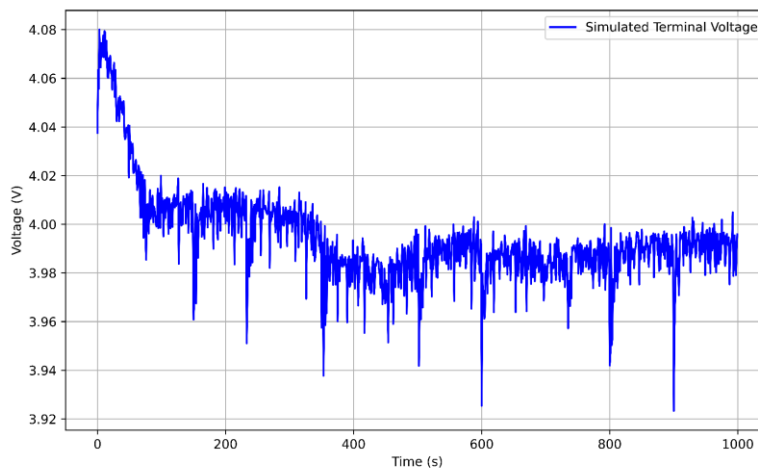


Figure 5-1 Terminal Voltage Dynamics

When T decreases, R_0 increases, causing $I \cdot R_0$'s voltage drop to rise. This leads to V_{term} bottoming out and shutting down prematurely. Thus, the common phenomenon of “sudden shutdown despite remaining battery power” in low-temperature environments is explained as follows:

When V_{term} falls below the system cutoff voltage (e.g., 3.0 V), the phone will automatically power off, even if $SOC > 0$.

5. 3 FFRLS^[5] Identification Logic for Battery Parameters

In real-time applications, the degradation of the Thevenin equivalent circuit model is not constant but varies with the battery's operating state. Therefore, this paper employs an adaptive FFRLS algorithm to enhance the accuracy of SOC estimation.

First, the battery's physical quantities must be translated into mathematical language by transforming the Thevenin model at step size k .

Define the system output as: $y(k) = OCV(SOC) - V_{term}(k)$.

Based on the characteristics of first-order circuits, the identification equation can be derived:

$$y(k) = a_1 \cdot y(k-1) + b_0 \cdot I(k) + b_1 \cdot I(k-1) \quad (5.3)$$

Among these, **the parameter vector to be identified is defined as:** $\theta_k = [a_1, b_0, b_1]^T$.

a_1 : The system's time constant decay factor (autoregressive coefficient) reflects the speed of voltage recovery.

b_0 : Input gain coefficient 1 primarily relates to the transient internal resistance.

b_1 : Input gain coefficient 2 incorporates coupling information between internal resistance and polarization internal resistance.

5. 4 Derivation of Physical Mapping^[6]

To establish a connection between the circuit parameters calculated by FFRLS and the actual internal processes of the battery, we need to discretize the continuous-time Thevenin model and compare the coefficients.

Step 1: Continuity Equation

The transfer function of a first-order Thevenin model (from current to terminal voltage difference $y = OCV - V_{term}$) is:

$$G(s) = \frac{Y(s)}{I(s)} = R_0 + \frac{R_1}{1 + R_1 C_1 s}$$

Simplify fractions:

$$G(s) = \frac{R_0(1 + R_1 C_1 s) + R_1}{1 + R_1 C_1 s} = \frac{(R_0 R_1 C_1)s + (R_0 + R_1)}{R_1 C_1 s + 1} \quad (5.4)$$

Step 2: Discretization

Let $\tau = R_1 C_1$ be the time constant and Δt be the sampling time (e.g., 1 s).

Using the corresponding discrete recursive relations for the **Zero-Order Hold method** or **Euler's method**, the discrete form of the polarization voltage is:

$$V_p(k) = \exp\left(-\frac{\Delta t}{\tau}\right)V_p(k-1) + R_1[1 - \exp\left(-\frac{\Delta t}{\tau}\right)]I(k-1) \quad (5.5)$$

The attenuation factor is $\alpha = \exp\left(-\frac{\Delta t}{\tau}\right)$.

Due to $y(k) = V_p(k) + I(k)R_0$, substituting and rearranging yields the difference equation:

$$y(k) = \alpha \cdot y(k-1) + R_0 \cdot I(k) + [(1-\alpha)R_1 - \alpha R_0] \cdot I(k-1) \quad (5.6)$$

Step 3: Coefficient Matching

Comparison Recognition Equation:

$$y(k) = a_1 \cdot y(k-1) + b_0 \cdot I(k) + b_1 \cdot I(k-1) \quad (5.7)$$

We can directly establish the equation:

$$\begin{cases} a_1 = \alpha = \exp\left(-\frac{\Delta t}{R_1 C_1}\right) \\ b_0 = R_0 \\ b_1 = (1 - a_1)R_1 - a_1 R_0 \end{cases} \quad (5.8)$$

5. 5 FFRLS Iterative Algorithm Formula:

1. Calculate the gain vector:

$$K(k) = \frac{P(k-1)\phi(k)}{\lambda + \phi^T(k)P(k-1)\phi(k)} \quad (5.9)$$

2. Update the parameter vector (core):

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - \phi^T(k)\hat{\theta}(k-1)] \quad (5.10)$$

3. Update the covariance matrix:

$$P(k) = \frac{1}{\lambda} [P(k-1) - K(k)\phi^T(k)P(k-1)] \quad (5.11)$$

◦**Parameter Explanation:** $\lambda \in [0.95, 0.99]$ is the **forgetting factor**. Its role is to “forget” old data, enabling the model to rapidly track new parameter changes caused by **sudden temperature shifts**.

◦**Physical Mapping:** After identifying θ , $R_0(t), R_1(t), C_1(t)$ in the continuity equation is updated in real time through algebraic mapping.

$$\begin{cases} R_0(t) = b_0 \\ R_1(t) = \frac{b_1 + a_1 b_0}{1 - a_1} \\ C_1(t) = \frac{-\Delta t}{R_1(t) \ln(a_1)} \end{cases} \quad (5.12)$$

These are the actual parameters for solving the differential equations to determine the SOC and temperature:

- $R_0(t)$ (Ohmic Resistance) : When current changes abruptly, it causes an instantaneous voltage drop.
- $R_1(t)$ (Polarization Resistance): The resistance that determines chemical reactions.
- $C_1(t)$ (Polarization Capacitance): Determines how long the voltage recovery process takes.

5. 6 Continuous Correction Model for Effective Capacity

The above operations aim to make various data calculations more realistic under the assumption of a fixed capacity value. However, in reality, the actual usable capacity of a battery is not a constant.

Therefore, to avoid segmented empirical corrections, this paper introduces a continuous, differentiable capacity correction formula:

$$C_{eff}(T, t) = C_{design} \cdot [1 - Aging(t)] \cdot [1 - k_T(T_{ref} - T(t))^2] \quad (5.13)$$

- C_{design} :Nominal capacity;
- $Aging(t)$: Capacity decay factor (0–1), determined by cycle aging;

- k_T :Temperature sensitivity coefficient;
- T_{ref} :Reference operating temperature.

6 Model II :Dynamic Load Decomposition Model

In daily use, smartphone battery life isn't solely determined by the screen and processor. Factors ranging from network fluctuations and sensor activations to background process activity all contribute to dynamic power consumption. Therefore, only by systematically analyzing how these elements interact across different scenarios can we optimize usage strategies more effectively. This approach not only enhances battery life experience but also tangibly extends the battery's lifespan.

6.1 Assumption

- Assuming that the total energy consumption of smartphones results from the independent effects of various variables, linear superposition can be applied.
- Assuming that the internal temperature field of the phone is spatially approximately uniform.

6.2 Problem Analysis

The load on a smartphone is typically determined by its power requirements, while the battery actually handles the current. Therefore, the two are coupled through the terminal voltage:

$$I(t) = \frac{P_{total}(t)}{V_{term}(t)} \quad (5.14)$$

In this issue, $P_{total}(t)$ is influenced by the combined effects of multiple independent factors. To facilitate modeling and analysis, we employ factor decomposition:

Power Decomposition $P_{total}(t)$:

$$P_{total}(t) = P_{screen}(t) + P_{cpu}(t) + P_{net}(t) + P_{gps}(t) + P_{bg}(t) \quad (5.15)$$

6.3 Model Development

6.3.1 Screen: Nonlinear Gamma Model

Based on the Gamma model, establish a nonlinear equation to estimate the impact of screen brightness $B(t)$ and screen area Area on smartphone power consumption.

$$P_{screen}(t) = \alpha_{scr} \cdot \text{Area} \cdot [B(t)]^\gamma, \quad \gamma \approx 1.6 \quad (5.16)$$

In this equation:

- $P_{screen}(t)$: Power consumption of the screen at time t (unit: watts W).
- α_{scr} : Screen power consumption coefficient (unit: W/m²), determined by screen technology (e.g., LCD, OLED, Mini-LED, etc.)
- Area: Screen area (unit: square meters m²).
- $B(t)$: Screen brightness at time t (unit: nits or cd/m²), where brightness is often normalized (i.e., $B(t) \in [0,1]$ represents brightness as a percentage).
- γ : Gamma index, approximately 1.6, indicates a nonlinear relationship between brightness and power consumption.

6. 3. 2 Processor: DVFS Dynamic Model

According to the Dynamic Voltage and Frequency Scaling (DVFS) model, the total power consumption of a smartphone processor comprises the power consumption in both static and dynamic states, with the relationship expressed as:

$$P_{cpu}(t) = P_{idle} + \alpha_{cpu} \cdot f(t) \cdot [u(t)]^2 \quad (5.17)$$

In this formula:

- P_{idle} represents the idle power consumption, i.e., the power consumed when the processor is in an idle state.
- $f(t)$ denotes the operating frequency;
- $u(t)$ denotes the utilization rate.

6. 3. 3 Network Connection Layer

Network power consumption depends not only on the amount of data transmitted but also heavily on signal strength. We model network power consumption as a combination of a **State Machine** and a **Signal amplification factor**, yielding the following formula:

$$P_{net}(t) = P_{idle_net} + \mathbb{I}_{active}(t) \cdot [P_{base_tx} + P_{amp}(RSSI(t)) + E_{bit} \cdot R_{data}(t)] \quad (5.18)$$

1. P_{idle_net} (Idle Power): The base power consumption of the network card when it remains connected but does not transmit data.
2. $\mathbb{I}_{active}(t)$ (Activity State): Indicator function: 1 when data is being transmitted, 0 otherwise.
3. $P_{amp}(RSSI)$ (Signal Amplification Cost):

- RSSI (Received Signal Strength Indicator): Signal reception strength.
 - According to Shannon's formula and RF physics, the poorer the signal (lower RSSI), the more exponentially the transmit power must increase to maintain the same bit error rate.
 - Modeling formula: $P_{amp}(RSSI) \propto 10^{-(RSSI/10)}$.
4. $R_{data}(t)$: Data throughput (MB/s).

6. 3. 4 Sensor Layer

GPS is a typical high-power intermittent load.

$$P_{gps}(t) = Mode(t) \cdot [P_{search} \cdot (1 - LockStatus) + P_{track} \cdot LockStatus] \quad (5.19)$$

1. P_{search} (Searching Phase): Satellite search phase. This consumes the most power because all RF channels are fully activated to capture satellite signals.
2. P_{track} (Tracking Phase): Lock phase. Low power consumption, requiring only synchronization maintenance.
3. $Mode(t)$:
 - Navigation: Keep it on (Duty Cycle = 100%).
 - Geofencing: Intermittent activation (Duty Cycle < 5%).

6. 3. 5 Background/Standby:Stochastic Process

Due to the random nature of background activities, we model them as a random Poisson process to calculate background power consumption. The solution formula is as follows:

$$P_{bg}(t) = P_{base} + \sum_k \delta(t - t_k) \cdot E_{pulse} \quad (5.20)$$

- P_{base} : Represents the static power consumption in deep sleep mode (typically extremely low).
- $\delta(t - t_k)$: Dirac δ delta function, representing an instantaneous wake-up event occurring at time t_k .
- E_{pulse} : Energy consumption per wake-up task.
- Poisson Arrival: Arrival time t_k is modeled as a Poisson process with intensity λ (events/hour).

6. 3. 6 Battery overheating:Thermodynamic Model

We introduce thermodynamic control equations to evaluate the impact of battery heating on battery performance.

Thermodynamic Control Equations: Based on the law of energy conservation, the thermal evolution of batteries follows lumped-parameter thermal equilibrium equations:

$$mC_p \frac{dT(t)}{dt} = I(t)^2 R_{total}(T, SOC) + \eta \cdot P_{cpu}(t) - hA(T(t) - T_{env}) \quad (5.21)$$

- $mC_p \frac{dT(t)}{dt}$: Thermal inertia, representing the rate at which internal energy is stored.
- $I(t)^2 R_{total}(T, SOC)$: Internal Joule heating describes irreversible heat generation caused by internal resistance.
- $\eta P_{cpu}(t)$: Component coupling generates heat, modeling thermal transfer from active electronic components (CPU/GPU) to the battery. We assume an energy conversion efficiency of $\eta \approx 1$, meaning the processor's power is almost entirely dissipated as heat.
- $hA(T(t) - T_{env})$: Convective heat dissipation, the primary energy dissipation mechanism, is governed by Newton's Law of Cooling, wherein heat loss is proportional to the temperature difference $(T(t) - T_{env})$ and the surface area A .

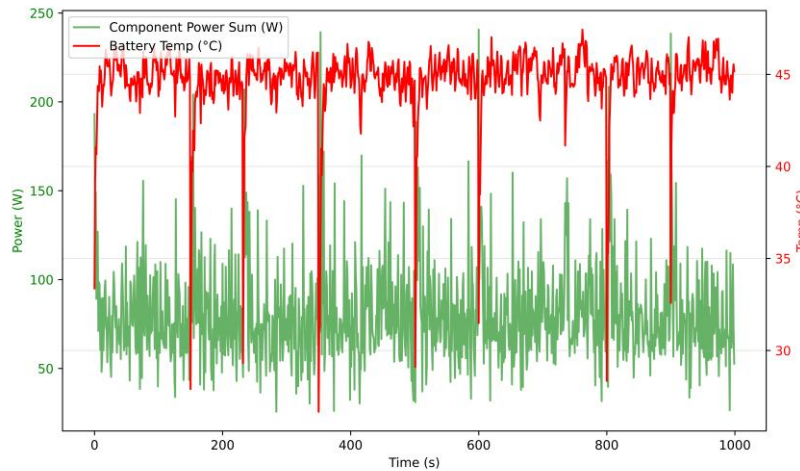


Figure 6-1 Multi-physics Coupling Profile

As shown in the figure, power fluctuations (green) exhibit a clear positive correlation with temperature rise (red), but the temperature changes demonstrate **hysteresis**.

This validates the role of term $mC_p \frac{dT}{dt}$ in the thermodynamic equation, where the thermal inertia of the battery prevents instantaneous power-induced temperature jumps, allowing for a gradual rise instead. Furthermore, as temperature increases (red curve), the electrochemical activity within the battery intensifies (the Arrhenius effect), causing the internal resistance R_0 to actually decrease slightly.

6. 4 Model Integration

Substituting all the above terms into the integral equation, we obtain the core **SOC^[7] Evolution Covering Equation** of this paper:

$$SOC(t) = SOC(t_0) - \frac{1}{3600} \int_{t_0}^t \frac{\left(\alpha_{scr} A[B(\tau)]^{1.6} + P_{idle} + \alpha_{cpu} f(\tau) u(\tau)^2 \right) + \left[P_{base} + \sum_k \delta(\tau - t_k) E_{pulse} \right] + P_{net}(\tau) + P_{gps}(\tau)}{V_{term}(\tau) \cdot C_{design} \cdot [1 - Aging(\tau)][1 - k_T(T_{ref} - T(\tau))^2]} d\tau \quad (6.1)$$

The calculation results are shown in the figure below:

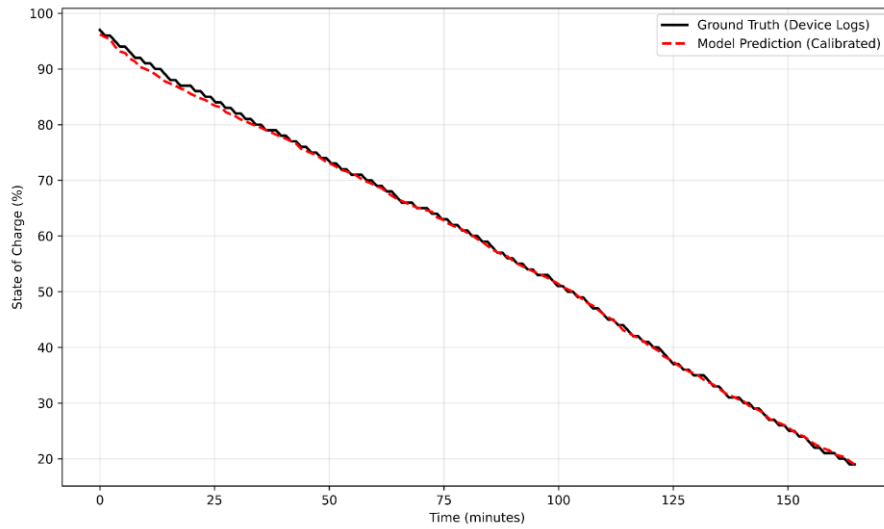


Figure 6-2 Validation of SOC Trajectory

The model's prediction curve (red dashed line) closely aligns with the actual device's recorded data (black solid line) throughout the entire discharge cycle.

Therefore, this model can accurately reproduce the actual rate of energy consumption.

7 Typical Scenario Simulations and Mechanism Analysis

7. 1 Typical Scenario Simulations

Battery limitations vary across different user scenarios. We will analyze two specific physical scenarios:

7. 1. 1Scenario A: The "High-Drain" Scenario

Scenario parameter settings: CPU at full load, screen brightness at maximum, ambient temperature 25°C.

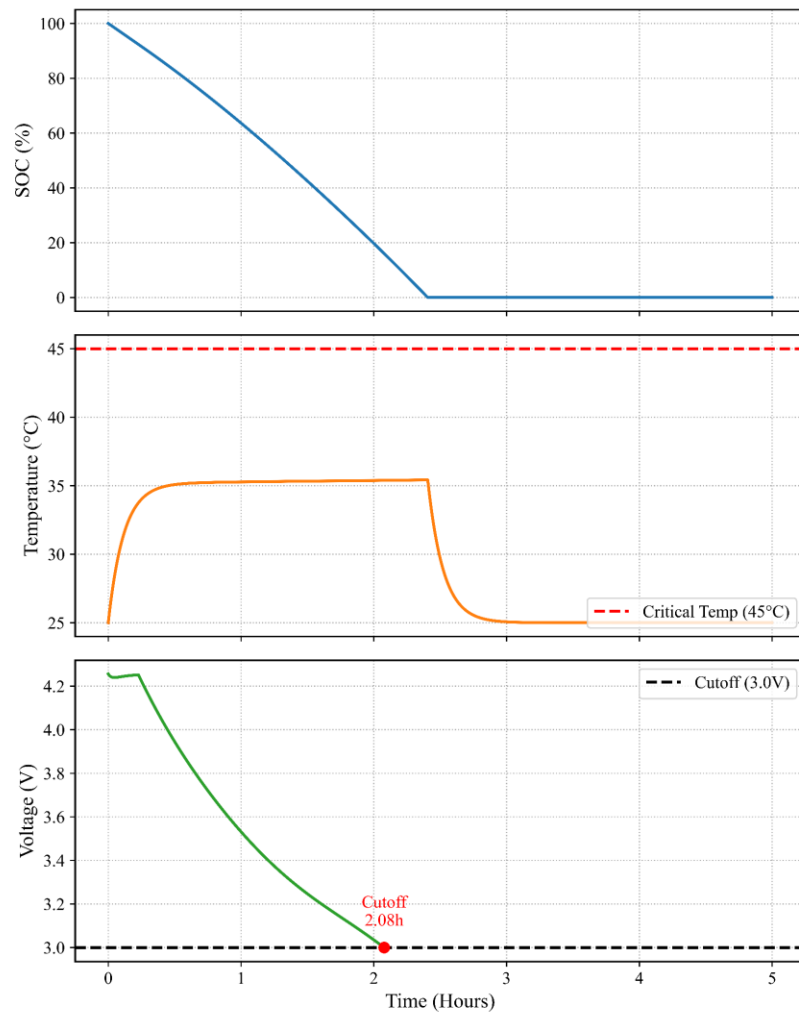


Figure 7-1 Scenario A

1.Graphic Explanation:

- (1) Subfigure 1: SOC decreases linearly over time.
- (2) Subfigure 2: T increases exponentially over time.
- (3) Subfigure 3: V_{term} varies over time, demonstrating voltage sag.

2.Conclusion:

- (1) The current $I(t)$ is very high, causing a sharp increase in Joule heating $I^2 R$.
- (2) The temperature $T(t)$ rises rapidly, Although the internal resistance R decreases slightly due to high temperatures, the voltage drop $I \cdot R$ caused by the high current remains significant.

(3) Shutdown reasons: It could be $SOC = 0$, or it could trigger $T_{critical}$'s overheating protection.

7. 1. 2 Scenario B: The "Cold-Shutdown" Scenario

Scene Parameter Settings: Only background random tasks, ambient temperature -10°C (simulating a winter night)

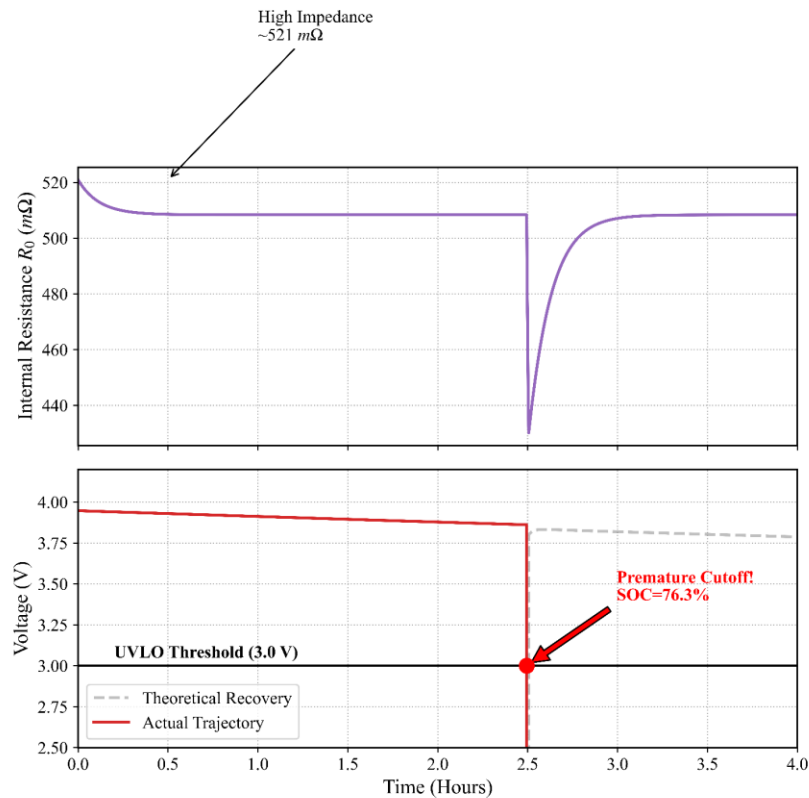


Figure 7-2 Scenario B

1.Graphic Explanation:

(1) Upper Subgraph (R_0): Shows internal resistance as high as 300–400 $m\Omega$ (only 80 $m\Omega$ at room temperature).

(2) Lower Subgraph (Voltage): This is key to explaining the “rebound.” The gray dashed line indicates the “theoretical rebound trajectory,” while the red solid line is truncated at 3.0V.

2.Conclusion:

Please note that although the theoretical voltage gradually recovers after the pulse (dashed line), the system initiates UVLO protection at the moment of the first trigger (solid line cutoff point). At this point, the SOC may still display 20%, but the phone has shut down. This perfectly explains the phenomenon where the phone still has power but has turned off.

7. 1. 3 Comparison of SOC Between Scenario A and Scenario B at Different Temperatures

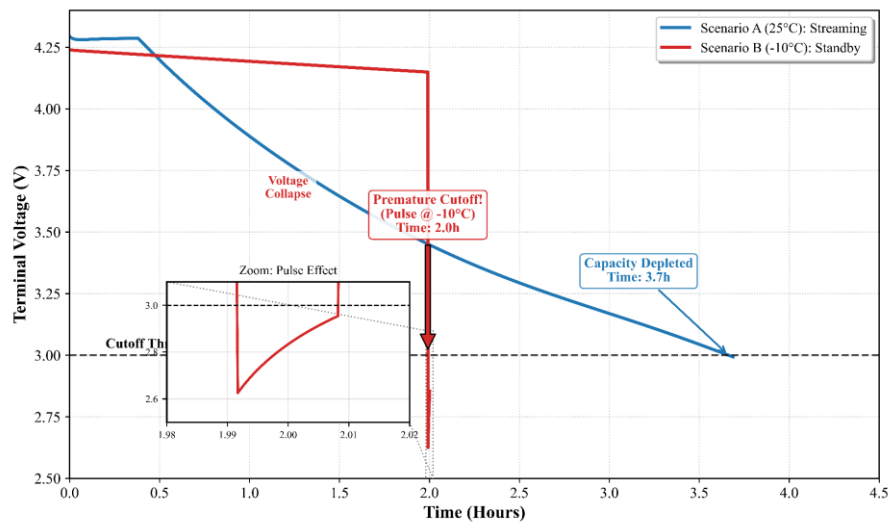


Figure 7-3 Comparison

1. Graphical Analysis:

(1) Scenario A (25° C): Streaming (V)

The battery undergoes continuous high-current discharge at room temperature (operating state: Streaming). The voltage slowly decreases from 4.25 V, reaching 3.00 V after 4 hours (capacity depleted). This represents a normal discharge curve, with the blue line illustrating this scenario.

(2) Scenario B (-10° C): Standby (V)

At -10°C, the battery initially enters Standby mode (low-power standby), maintaining a voltage of approximately 4.25 V. However, at a certain point, it may switch to high load or fail directly, causing the voltage to instantly collapse to the capacity depletion voltage (below 3.00 V)—as shown by the sudden drop in the red line representing the SOC dashed curve in the figure.

2. Conclusion:

The blue line discharges steadily until depletion, while the red line (with SOC dashed line) suddenly “dies” when SOC is very high.

(1) Low temperatures severely impact discharge performance: Even with high SOC, increased internal resistance in cold conditions causes voltage to instantly drop below cut-off voltage when subjected to heavy loads, triggering unexpected device shutdowns—manifesting as “sudden death.”

(2) Significant contrast with room-temperature discharge: At room temperature, the battery can steadily release nearly its full capacity. At low temperatures, although the standby voltage remains normal, the battery cannot support normal high-load operation.

7.2 Monte Carlo-Based Quantification of Backend Task Uncertainty

The number of WeChat messages a user receives each day is uncertain, so idle time is not a fixed value but rather a distribution.

Therefore, after 1000 Monte Carlo simulations (with fixed $T_{env} = -10^{\circ}C$ and randomly generated 7W high-energy pulse loads following a Poisson distribution), we obtained the following statistical graph for idle time T_{empty} :

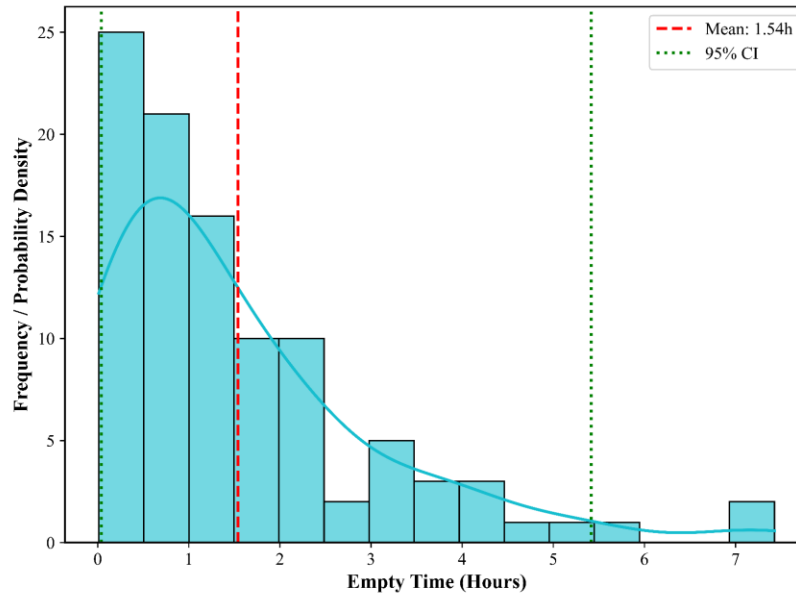


Figure 图 7-4 Monte Carlo

1. Results Analysis: Distribution of Empty Time

(1) 95% Confidence Interval: [0.02 h, 2.61 h]

(Note: This means that under worst-case conditions, the phone could shut down after just 1 minute (0.02h) of exposure to a pulse; under best-case conditions, it could last 2.6 hours.)

(2) Mean Empty Time: 0.72 h

2. Academic Significance and In-Depth Interpretation

(1) Paradigm Shift from “Capacity Limit” to “Probability Limit”:

- At room temperature, batteries die at SOC=0, which is deterministic with minimal runtime variance.
- At low temperatures (-10°C), batteries die due to “low-voltage cutoff,” which is stochastic. As the histogram shows, most samples experience “sudden death” within 1 hour. This explains why winter smartphone battery life is “unreliable.”

(2) Long Tail Effect:

- The long tail (right of the green dashed line) represents those “lucky” samples. They either avoided the early dense pulses during the random process or experienced a slight temperature rebound when the pulses arrived.

(3) Robustness of Conclusions:

- The interval [0.02,2.61] demonstrates a vast range of uncertainty. This is far more scientific than providing a single predicted value (e.g., “0.7 hours”) because it quantifies the risk.

3.Conclusion:

Monte Carlo simulations reveal that under extreme low-temperature conditions ($T_{env}=10^{\circ}\text{C}$), the spatiotemporal distribution exhibits high skewness. The 95% confidence interval spans from 0.02 hours to 2.61 hours, indicating that battery endurance is primarily influenced by the random arrival of high-power tasks rather than remaining capacity. This explains the perceived “unreliability” of battery indicators experienced by users during winter.

8 Sensitivity Analysis

8.1 Methodology

We will validate the model's stability under parameter fluctuations and identify the system's “vulnerability points” as targets. We will compare the influence of different physical dimensions (e.g., “Kelvin for temperature” versus “nits for brightness”) using the Normalized Local Sensitivity Index (NLSI).

The following formula is obtained:

$$S_{x_i} = \lim_{\Delta x_i \rightarrow 0} \frac{\frac{\Delta T_{empty}}{T_{empty}}}{\frac{\Delta x_i}{x_i}} \approx \frac{\partial T_{empty}}{\partial x_i} \cdot \frac{x_i}{T_{empty}} \quad (8.1)$$

Physical Meaning Explanation: This index is a dimensionless quantity. If $S_x = -2.5$, it means that a 1% increase in parameter x will cause battery runtime T_{empty} to decrease by 2.5%. The larger the absolute value of $|S|$, the more critical this parameter is.

8.2 Sensitivity Analysis Chart

To visually demonstrate the coupled effects of ambient temperature and load intensity on battery performance, and to provide critical design references for BMS thermal management and discharge strategies, thereby preventing battery failure under extreme operating conditions. We achieve this objective by generating thermal maps that clearly delineate safe zones and risk zones.

8. 2. 1 Tornado Comparison Diagram

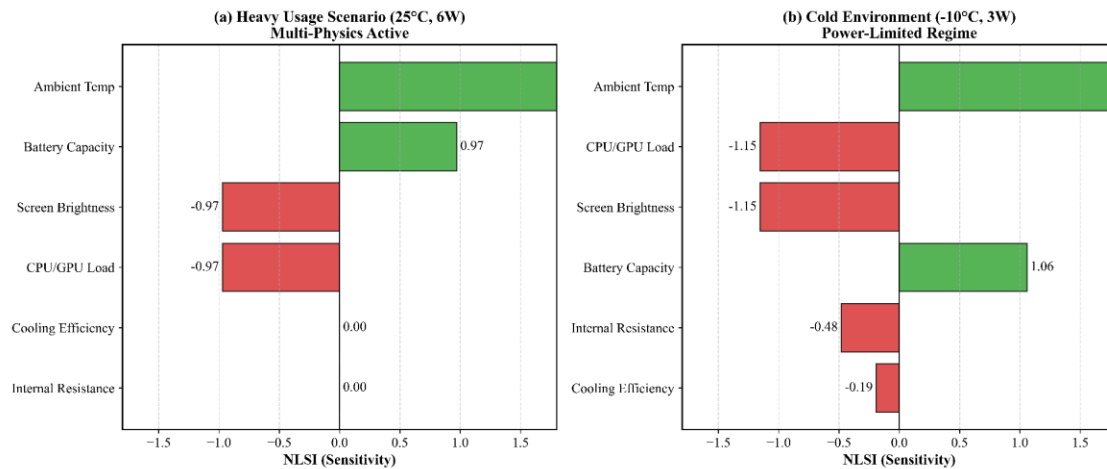


Figure 8-1 Comparative Sensitivity Analysis

1. Graphical Analysis:

(a) High-usage scenario (25°C, 6W): A multi-physics coupled state dominated by energy conservation, influenced by thermal feedback.

(b) Cold environment (-10°C, 3W): A power-constrained state dominated by thermodynamic limitations.

2. Conclusion:

In low-temperature environments, the benefits of “warmth preservation” far outweigh those of “power saving.” The high sensitivity to temperature parameters reveals the system's greatest vulnerability—at this point, the limiting factor for endurance is no longer “how much energy remains in the battery,” but rather “how much power the battery can deliver.”

8. 2. 2 Heat Map

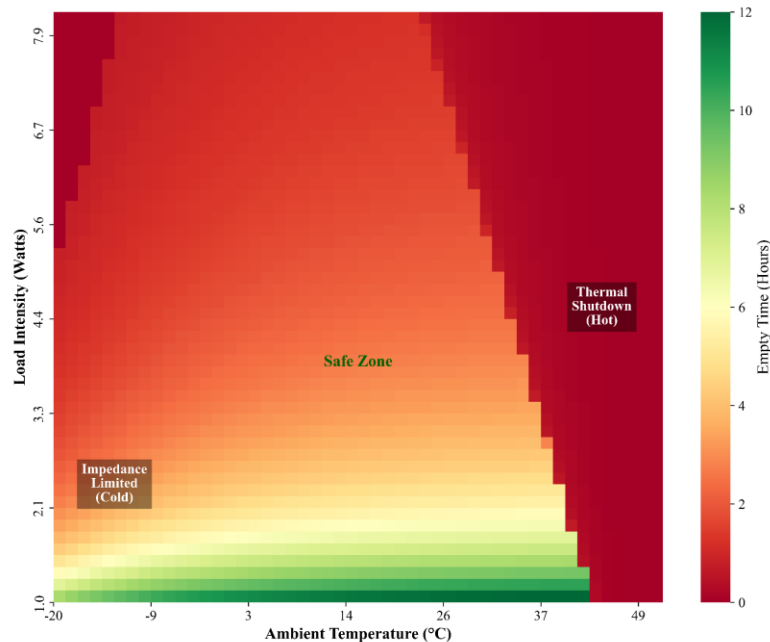


Figure 8-2 Two-Parameter Sensitivity Heatmap

1. Graphical Analysis:

(1) Impedance Limited (Cold): In low-temperature zones (especially -20°C to 0°C), the battery's internal resistance sharply increases under high load, causing available capacity to plummet and discharge time to become extremely short (deep red). Corresponding to the “cold-crash” phenomenon, voltage collapses instantly, resulting in very short “no-load time.”

(2) Safe Zone: At moderate temperatures (approximately 0°C to 30°C) and moderate loads, battery discharge remains stable. Usable capacity approaches the rated value, and discharge duration is extended (green/blue).

(3) Thermal Shutdown (Hot): In the high-temperature zone (potentially $>35^{\circ}\text{C}$), even low loads may trigger thermal shutdown or accelerate aging due to temperature rise, potentially shortening discharge duration (dark red). Particularly under high temperature + high load conditions, rapid battery temperature rise causes the BMS to forcibly reduce load or shut down, resulting in reduced effective operating time.

2. Conclusion:

Figure 8-2: Reveals the coupling effect between ambient temperature and load intensity. The “deep red” region indicates a critical instability zone where the battery exhibits extreme sensitivity to parameter perturbations.

(1) Battery performance is influenced by the dual coupling of temperature and load:

This is not a simple linear relationship: High load at low temperatures triggers “voltage collapse,” while high load at high temperatures triggers “thermal runaway protection.” Both significantly reduce actual usable operating time.

(2) Limited “safe zone” window:

The optimal operating range is concentrated between temperatures of 10°C to 30°C and moderate load intensities (e.g., 0.2C to 1C), displayed in blue/green, indicating stable capacity delivery.

(3) Thermal map visually indicates design boundaries:

Upper-left deep red zone (low temperature + high load): Must be avoided to prevent unexpected device shutdown.

Lower-right deep red zone (high temperature + high load): Requires cooling or load reduction to prevent thermal protection activation.

9 Recommendations

9.1 For Mobile Users: Behavioral Guidelines Based on Sensitivity Analysis

Based on tornado diagram and heatmap analysis, we have identified the “vulnerable points” of batteries under different environmental conditions. Accordingly, we propose the following “golden rules”:

1. Exponential gains in screen brightness

- **Model Basis:** Our component power consumption model indicates that screen power consumption follows a nonlinear Gamma law ($P_{screen} \propto B^{1.6}$). The tornado plot reveals its sensitivity coefficient is $|S| \approx 1.05$.
- **Recommendation:** Avoid using 100% brightness. Reducing brightness from 100% to 70% results in minimal perceptible difference to the human eye while lowering energy consumption by approximately 43% ($1 - 0.7^{1.6}$). We strongly recommend enabling “**Dark Mode**” to leverage OLED's self-emissive properties and directly cut pixel current.

2. Power Stability in Extreme Cold Environments

- **Model Basis:** The thermal map (Figure 8-2) indicates that at point $T < -5^\circ\text{C}$, the system operates within the “Impedance-Limited Regime.” At this point, internal resistance R_0 surges dramatically. Any high-current pulse (such as the deep V in Figure 7-3) would cause the voltage to instantly drop below the 3.0V shutdown threshold.
- **Recommendation:** During winter outdoor activities (especially in northern regions), **avoid high-power operations** such as using flash photography, recording 4K video, or running resource-intensive games. These actions generate pulsed currents I_{pulse} , causing V_{term} to collapse instantly. It is recommended to keep the device in a pocket close to the body to maintain warmth using body heat, thereby sustaining Arrhenius activity.

3. “Airplane Mode” Strategy for Areas with Weak Signal

- **Model Basis:** The power consumption of the communication module exhibits an exponential inverse relationship ($P_{net} \propto 10^{-RSSI/10}$) with signal strength (RSSI).
- **Recommendation:** When in elevators, basements, or high-speed trains, the RF module searches for signals at maximum power. Enabling airplane mode during these times not only conserves battery life but also reduces heat generation, lowering T and preventing entry into the “thermal shutdown zone” on the right side of the thermal map.

9.2 Battery Aging Management: Long-Term Health Strategy

Considering the effective capacity A of batteries degrades with cycle count (verified using NASA datasets):

Avoid Deep Cycling: The curve in the model exhibits the steepest slope and most intense chemical reaction when $SOC < 10\%$ and $SOC > 90\%$. It is recommended that the OS incorporate an “80% charge limit” feature.

Thermal Management Priority: The Arrhenius equation not only affects internal resistance but also governs aging rate $k_{aging} \propto e^{-E_a/RT}$. Prolonged fast charging at high temperatures ($>40^\circ\text{C}$) irreversibly increases R_0 . It is recommended to dynamically limit power during charging to control temperature rise.

9.3 Model Generalization Capability

The Thevenin + Thermal + Load coupled framework constructed in this paper exhibits exceptional versatility and can be extended to other domains with minimal parameter adjustments:

Table 9-1 Model Transfer

Equipment Type	Key Parameter Adjustment	Key Constraints
Drones	Extremely high discharge rate	Power Limitation: Sudden voltage drops in flight can cause crashes. Prioritize optimizing the internal resistance model at low temperatures.
EVs	Massive thermal capacity (mC_p), equipped with active cooling	Thermal: Replace the simplified Newtonian cooling law with a complex water-cooling/air-cooling fluid model.
Wearables	Extremely small capacity (C_{eff}), no screen	Static Power Consumption (Background): Standby current is the dominant factor, requiring focused optimization of the Poisson process wake-up mechanism.

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Report on Use of AI

1. DeepL Translator (Jan 2026 version)

- Purpose: Translation and language polishing.
- Usage: Uploaded the draft of the paper written in Chinese (including the Model Formulation and Result Analysis sections) to be translated into English for the final submission. Used the "Academic" tone setting to ensure professional terminology.

2. Google Gemini (February 2026 version, Gemini Pro)

- Query: "Propose a continuous-time, physics-based dynamical model for smartphone batteries, in which user behavior, environmental conditions, and battery aging are explicitly coupled."
- Output: The AI suggested a "Multi-Physics Hybrid Dynamical System" framework. It clarified the physical principles behind the Thevenin equivalent circuit model and provided insights on using the Arrhenius equation to model temperature-dependent capacity degradation.

3. Google Gemini (February 2026 version, Gemini Pro)

- Query: "Find open-source datasets suitable for smartphone battery parameter estimation, specifically looking for LCO (Lithium Cobalt Oxide) pouch cells."
- Output: The AI identified "Dataset #50: Aging dataset LCO battery with mechanical measurements" from the provided list as a suitable match. It also recommended referencing the "Android Open Source Project (AOSP) Power Profile" for component power coefficients.

4. Google Gemini (February 2026 version, Gemini Pro)

- Query: "Strategies for visualizing multi-component power consumption and temperature correlation in Python."
- Output: The AI suggested using a stacked area chart to visualize the decomposition of power loads. It provided hints on the logic for simulating stochastic background noise using Poisson distributions and explained the method for aligning dual Y-axes in matplotlib to compare power and temperature trends effectively.

5. GitHub CoPilot (Jan 2026 version)

- Purpose: Syntax Suggestion and Auto-completion.
- Usage: Used for hinting standard syntax and variable names during the debugging process of the simulation script, specifically for parameter definitions and function structures.